

HumanConQuad: Human Motion Control of Quadrupedal Robots using Deep Reinforcement Learning

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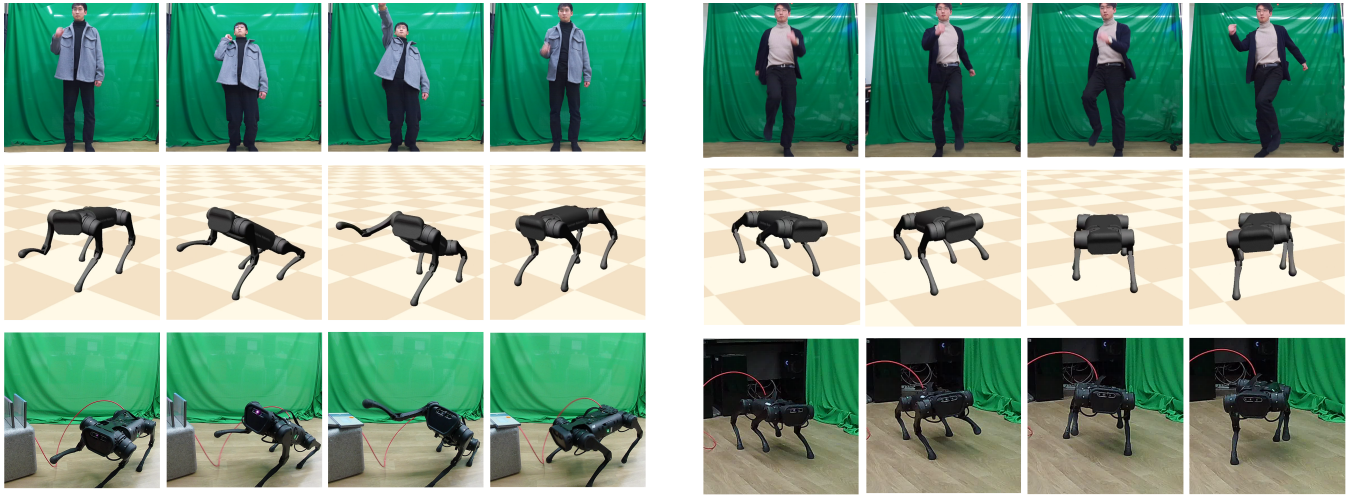


Figure 1: Our interface allows users to control quadrupedal robots to perform various tasks using human motion.

ABSTRACT

Robotic creatures are capable of entering hazardous environments instead of human workers, but it is challenging to develop a fully autonomous agent that can work independently in unstructured scenes. We propose a human motion-based control interface for quadrupedal robots that promises adaptable robot operations by reflecting the user’s intuition directly to the robot’s movements. Designing motion interface for different morphologies conveys tricky problems in solving dynamics and control strategies. We first retarget the captured human motion into the corresponding robot’s kinematic space with proper semantics using supervised

learning and post-processing techniques. Second, we build the motion imitation controller to track the given retargeted motion using deep reinforcement learning with task-based curriculums. Finally, we apply domain randomization during training for real-world deployment. (Video¹)

CCS CONCEPTS

• Computing methodologies → Robotic planning; Physical simulation.

KEYWORDS

HumanConQuad, Physcis Simulation, Human Motion-Based Control, Hetero-Morphological Control, Deep Reinforcement Learning

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¹Supplementary Video: <https://sites.google.com/view/humanconquad>

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1 INTRODUCTION

The legged robots entering dangerous environments bring safety to human workers through their motor capabilities. Many robotists develop control strategies to follow this virtue. Traditional mathematical approaches such as trajectory optimization or model predictive control (MPC) show the ability to generate robust robot motion but often suffer from sophisticated dynamics model [Raibert 1984]. Reinforcement learning (RL) controllers alleviate calculations dealing with model complexity in the physics simulation. However, it is still challenging to develop autonomous agents that can work in various unstructured scenarios.

Using human motion as a control input permits users to reflect the intention directly. Therefore, human posture-based control has been widely studied in the fields of computer animation and robotics from human to humanoid control interface [Choi et al. 2021] and even from human to non-humanoid characters [Yamane et al. 2010].

We design a novel human motion interface for a quadrupedal robot that promises flexible operations. We develop an effective motion retargeting module using supervised learning with post-processing of contact and temporal consistency corrections to deal with the ambiguity of the mapping function. Then We propose a motion imitation policy with curriculum and a hierarchical structure for the user to execute various motor tasks seamlessly on simulated and real robots.

2 MOTION RETARGETING

The first module of our interface retargets the human user's motion q into the corresponding robot motion p . We introduce supervised learning as a retargeting method.

We prepare the dataset $\mathcal{D}$ for learning the retargeting network by collecting matching pairs of human and robot motions. The user first generates the robot motion clips for a target task. Then, we collect the matching human motion sequences by asking a human user to act the "corresponding motions" while showing the generated robot motion clips. Consequently, this approach provides the user's intuitive mapping into the retargeting function.

We conduct our retargeting function using a multi-layer perceptron (MLP) from the given dataset $\mathcal{D}$. Our MLP consists of three leaky ReLU layers and one final hyperbolic tangent layer. Our loss function aims to minimize the difference between output and the target value between root orientation, joint angles, and end-effectors position and velocity.

The learned function often generates physically invalid motions in practice. This inconsistency often causes control failure. We correct the contact and temporal consistency as a post-processing of the retargeting module. We fix the supporting leg poses to prevent unnecessary foot sliding. The abrupt joint motion is clipped with the velocity limit for temporal correction.

3 MOTION IMITATION

We build our second module to build a control policy to imitate the given kinematic reference motion. We employ the motion imitation learning framework similar to Peng et al. [2018]. Our control policy

takes the generated robot trajectories for training a retargeting function in the first module. Our RL controller takes the robot sensor data, previous actions, and reference poses as the observation and outputs the PD target for the twelve joint motors of a robot. The reward function encourages the agent to imitate the given reference motion while adapting to the physics simulation. The policies are optimized with Proximal Policy Optimization.

The naive learning often causes policy to produce conservative motion which is stuck in a steady motion to only focus on balancing. To solve this issue, we introduce two curriculum strategies while learning: a task type as a primary and difficulty of the task as a secondary. For a task-type curriculum, we train simple tasks first and composite tasks later. For the task difficulties curriculum, we first train with the narrow motion range for the task and expand to wider workspaces.

The real world and the simulation environment show the dynamics difference which often causes the failure of deployment from the policies trained in the simulation environment to the real world. We utilize Domain Randomization which randomizes simulation parameters during the training for our real-world deployment.

4 RESULT

We evaluate our framework from three perspectives. First, we demonstrated control tasks in both simulated and real environments. The task includes various single tasks such as tilting, manipulation, and walking and further composite tasks which can achieve complex missions. Second, we conduct an ablation study to prove the effectiveness of important components of our system. We show that our consistency correction, curriculum learning, and domain randomization are essential to acquiring the control performance. Finally, we build a qualitative comparison of our system with previous motion-based interfaces. Our system outperforms the previous works by providing real-time, flexible, and real-world control (Figure 1).

5 CONCLUSION

We presented a human motion control interface for quadrupedal robots using motion capture with two main components: motion retargeting module and control policy. Still, there are rooms for our system to be improved. First, Kinect's delay and instability occasionally cause a system failure so a motion capture system with more stability will be more suitable. Moreover, we can expand this study to predict user intentions from history to improve the motion quality.

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